Lot Type Expansion — Considerations

Overview

The Market Value API currently trains and scores **Yearling** lots only. All sales in the system are filtered to `salesLotTypeName = 'Yearling'` across training (`train_model.py`), feature building (`run_rebuild.py`), and scoring (`score_sale.py`).

The database contains 11 lot types across 10+ countries, but only a subset have enough sold data (with prices) to support model training. This document evaluates each lot type for feasibility, outlines a phased expansion roadmap starting with **Weanling**, and identifies technical challenges that must be addressed.

Current Pipeline (Yearling Only)

1. **Training** — `fetch_sale_lots()` and `fetch_hist_lots()` filter on `salesLotTypeName = 'Yearling'`
2. **Feature building** — Sire/dam/vendor metrics computed from yearling-only historical data
3. **Session median** — Computed from sold yearling lots in the sale (or prior year yearling fallback)
4. **Scoring** — Model predicts `log_price_index = log(hammer_price / session_median_price)` for yearling lots
5. **Write-back** — Predictions stored in `tblHorseAnalytics` keyed by horse ID and sale ID

Expansion Goal

Support multiple lot types with minimal pipeline changes. Priority order:

1. **Weanling** — Similar to yearling, same features, good data volume
2. **Broodmare** — Different price drivers, needs separate feature engineering
3. **Breeze Up** — Small dataset, may need workout data
4. **Race Horse** — Requires race performance data, fundamentally different model

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Data Volume (since 2020)

Lot Type	Total Lots	Sold (with price)	Clearance	Priority
Yearling	154,814	82,272	53%	**Current**
Broodmare	25,738	13,449	52%	Medium
Weanling	31,069	10,894	35%	**High**
Race Horse	22,489	6,124	27%	Low
Breeze Up	12,762	2,439	19%	Low
Horse In Training	8,525	501	6%	Not viable
Stallion/Prospect	146	34	23%	Not viable
Stallion Share/Nom	46	26	57%	Not viable

Lot Type	Total Lots	Sold (with price)	Clearance	Priority
Foal at Foot	38	20	53%	Not viable
Broodmare in Foal	814	0	0%	Not viable
Breeding Right	2	0	0%	Not viable

Phase 1: Weanling (Recommended Next)

Why: Most similar to yearling — same sire/dam/vendor features apply, good data volume (~11k sold).

Option A: Separate Weanling Model

Train a dedicated model using only weanling sale data.

Pros:

- Model fully tuned to weanling pricing patterns
- Independent calibration offsets and elite scaling thresholds
- No risk of yearling data biasing weanling predictions
- Can evolve independently (different features, thresholds)

Cons:

- Less training data (~11k sold vs ~82k yearlings) — may underfit
- Additional model to maintain per region (e.g. `aus_v7_weanling`)
- Duplicate pipeline code or config to manage two model types
- Sire/dam/vendor metrics computed separately — smaller sample sizes

Changes required:

- `config.json` — Add `weanling_model` field per region (e.g. `"weanling_model": "aus_weanling_v1"`)
- `src/run_rebuild.py` — Parameterize lot type filter, separate session median per lot type
- `src/train_model.py` — Accept `lot_type` parameter, filter training data accordingly
- `src/score_sale.py` — Select model based on lot type
- `api.py` — Training and scoring endpoints accept lot type parameter
- New model directories per region (e.g. `models/aus_weanling_v1/`)

Option B: Combined Model with Lot Type Feature

Include weanling lots in the existing training pipeline alongside yearlings, with a `lot_type` encoded feature.

Pros:

- More training data overall (~93k combined) — model learns shared patterns
- Simpler to maintain — one model per region
- `lot_type` feature lets the model learn a baseline price offset between the two
- Shared sire/dam/vendor metrics have larger sample sizes

- If accuracy isn't sufficient, can split into separate models later (Option A)

Cons:

- Yearling data dominates (~88% of training set) — weanling signal may be diluted
- Single calibration offsets and elite scaling — may not be optimal for both
- Session median mixing could skew predictions if not handled per lot type

Changes required:

- `src/run_rebuild.py` — Expand filter to include weanlings, add `lot_type` column
- `src/train_model.py` — Include `lot_type_encoded` in feature columns
- `src/score_sale.py` — Add `lot_type_encoded` to scoring features
- Session median logic — Compute per lot type within a sale

IF Start with Option B, evaluate:

1. **Add weanling lots to training data** — Change `salesLotTypeName = 'Yearling'` filter to `IN ('Yearling', 'Weanling')`
2. **Add `lot_type` feature** — Encode as `lot_type_encoded` (0 = Yearling, 1 = Weanling) in training and scoring
3. **Session median per lot type** — Compute separate session medians for yearling and weanling lots within the same sale to avoid cross-contamination
4. **Train and evaluate** — Compare MAPE and quantile coverage for yearlings (should stay the same) and weanlings separately
5. **If weanling accuracy is poor** — Switch to Option A with a dedicated weanling model

Considerations:

- Weanlings typically sell for less than yearlings (less physical maturity, further from racing)
- Same sire/dam/vendor metrics are relevant for both
- A Mixed Sale (type 6) can contain both yearling and weanling lots — the `lot_type` feature handles this correctly
- Elite scaling thresholds may need separate values per lot type in future

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Technical Challenges for Adding Lot Types

These apply to weanlings (Phase 1) and become more significant for other lot types.

1. Session Median Contamination

`compute_sale_median()` currently calculates one median per sale. A Mixed Sale (type 6) with both yearlings and weanlings would blend their prices into one median, skewing the `log_price_index` target variable for both lot types. **Fix:** Compute session median per lot type within each sale.

2. Sire/Dam/Vendor Metrics Are Lot-Type Blind

`fetch_hist_lots()` in `run_rebuild.py` only pulls yearling lots for computing sire, dam, and vendor metrics. Questions to resolve:

- Should weanling sire stats be computed from weanling sales only, or combined yearling+weanling?
- A sire with 50 yearling sales but 3 weanling sales — which sample to use?
- Combined gives larger sample sizes but may dilute lot-type-specific pricing signals
- **Recommendation:** Start with combined metrics (simpler), evaluate if separate metrics improve accuracy

3. Prior Year Median Fallback

`fetch_prior_year_median()` in `run_rebuild.py` hardcodes `salesLotTypeName = 'Yearling'`. When scoring a weanling sale with no sold lots, this falls back to the yearling median from prior year — wrong baseline. **Fix:** Pass lot type through and filter accordingly.

4. Elite Scaling Thresholds

`elite_scaling.threshold` in `config.json` is tuned for yearling prices (e.g. AUS = $500,000). Weanlings sell for significantly less — this threshold would rarely trigger, making elite scaling ineffective for weanlings. **Fix:** Per-lot-type thresholds in config, or scale threshold proportionally based on typical lot type price ratio.

5. Calibration Offsets

`calibration_offsets.json` per model is calibrated on yearling residuals. Adding weanlings changes the error distribution, potentially degrading calibration for both. **Fix:** Recalibrate after retraining, or maintain per-lot-type offsets.

6. Confidence Tiers

`close_threshold` and `extreme_threshold` for confidence scoring are yearling-calibrated. Weanling price distributions may be tighter or wider. **Fix:** Evaluate confidence tier accuracy per lot type after retraining; adjust thresholds if needed.

7. Database Write-Back

`upsert_to_database()` writes predictions to `tblHorseAnalytics`. The same horse could appear as a weanling in one sale and a yearling in another. Need to verify the table schema handles multiple predictions per horse across different lot types without overwriting.

8. Frontend Impact

The UI currently assumes yearling-only data. Areas to update:

- Sale detail view — display lot type per lot
- Scoring display — may need lot type indicator
- Filtering — allow filtering by lot type
- Sale search — lot type counts in search results

Phase 2: Broodmare (Future)

Why: Second largest sold dataset (13.4k), commercially important.

Challenges — different price drivers:

- Breeding record (foals produced, their race results)
- In-foal status and covering sire quality
- Age matters differently (peak breeding years vs too old)
- Pedigree page matters more (stakes-producing mares)

Approach: Likely needs a **separate model** with different feature engineering:

- Dam racing record (earnings, stakes wins)
- Progeny performance (winners, stakes winners)
- Covering sire (if in foal)
- Age and breeding history
- Sire/broodmare sire quality metrics

Data availability: Need to verify that progeny performance and covering sire data are accessible in the database.

Phase 3: Breeze Up (Future)

Why: 2.4k sold lots, growing sale category.

Challenges:

- Workout/breeze times are a key price driver (not currently in the database?)
- Physical assessment plays a bigger role than yearlings
- Smaller dataset — may need to combine with yearling model

Approach: Could potentially extend the yearling model with:

- `lot_type` feature (as with weanling)
- Breeze time features if available
- Otherwise, sire/dam/vendor features still apply as a baseline

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Phase 4: Race Horse / Horse In Training (Future — Low Priority)

Why: 6.1k sold racehorses, but very different valuation model.

Challenges:

- Price driven by race record (earnings, wins, ratings)
- Current form and soundness
- Age and distance preferences
- Training status and trainer reputation
- Very low clearance for HIT (6%) — most data is unsold

Approach: Would require a completely separate model and feature pipeline with access to race performance data.

Not Viable

The following lot types have insufficient sold data for meaningful model training:

- **Horse In Training** — Only 501 sold (6% clearance)
- **Stallion/Stallion Prospect** — 34 sold
- **Stallion Share/Nomination** — 26 sold
- **Foal at Foot** — 20 sold
- **Broodmare in Foal** — 0 sold (no price data)
- **Breeding Right** — 0 sold

Appendix: Data Volume by Country (since 2020)

Yearling — by Country

Country	Total	Sold	Clearance
USA	56,054	33,924	70%
AUS	39,859	28,125	84%
GBR	19,440	4,008	21%
JPN	12,197	3,752	32%
NZL	9,915	6,489	76%
FRA	9,472	3,176	34%
IRE	3,877	2,798	77%
ZAF	2,598	0	—
GER	953	0	—
CAN	449	0	—

Weanling — by Country

Country	Total	Sold	Clearance
USA	13,005	5,530	53%
AUS	7,155	4,362	76%
IRE	4,780	0	—
GBR	3,132	88	3%
NZL	994	674	77%
JPN	803	180	23%
FRA	747	60	8%

Country	Total	Sold	Clearance
GER	453	0	—

Broodmare — by Country

Country	Total	Sold	Clearance
USA	15,797	9,994	77%
AUS	4,557	3,058	77%
GBR	3,278	1	0%
JPN	1,139	358	32%
IRE	876	0	—
NZL	70	36	55%
FRA	20	2	10%
GER	1	0	—

Race Horse — by Country

Country	Total	Sold	Clearance
USA	8,571	3,298	53%
GBR	7,374	1,091	16%
FRA	4,097	226	6%
AUS	2,010	1,453	84%
GER	193	0	—
IRE	133	0	—
NZL	70	36	55%
HKG	41	20	49%

Breeze Up — by Country

Country	Total	Sold	Clearance
USA	4,086	637	17%
GBR	2,972	0	—
AUS	2,386	1,323	74%
FRA	775	0	—
ZAF	759	0	—

Country	Total	Sold	Clearance
IRE	705	0	—
NZL	640	404	81%
JPN	439	75	17%

Horse In Training — by Country

Country	Total	Sold	Clearance
GBR	4,155	12	0%
USA	3,520	362	11%
JPN	396	0	—
IRE	286	0	—
AUS	165	127	91%
GER	2	0	—
FRA	1	0	—